Requirements Gathering

1- Stakeholders Analysis

Stakeholder	Role	Need from Analysis
Pharmacovigilance Officer	Monitors drug safety	Most frequent reactions and serious / death rates per drug and age group
Drug Development / Clinical Research Manager	Tracks the trial pipeline	Number, phase and status of trials by drug and sponsor investigational drugs
Financial Analyst / Investor	Evaluates companies	Compare LLY VS NVO and relate performance to public interest and approval events
Public Health Awareness Officer	Educates the public	Common reactions, which groups report most, where interest is rising
Researcher / Student	Studies the field	An interactive overview of the seven drugs with filtering

2- User Stories & Use Cases

User Stories

As a Pharmacovigilance Officer, I want to compare series and death rates across drugs, so that I can identify drugs needing closure monitoring.

As a Clinical Research Manager, I want to see trial counts by phase, status and sponsor, so that I can track Novo Nordisk and Eli Lilly pipeline activity.

As a Financial Analyst, I want to compare LLY and NVO prices next to search interest for Ozempic and Mounjaro, so that I can understand how public interest affects the market.

As an Awareness Officer, I want to see reports by age, sex and country, so that I can target awareness campaigns at the right groups.

As a Researcher, I want to filter every page by drug and year, so that I can focus on a specific drug or period.

User Cases

Compare drug seriousness
1) Open the Drug Safety page.
2) Select years in the year filter.
3) View serious and death rates per drug.

Explore a drug's reactions
1) Select a drug.
2) View the top 10 reactions.

Follow clinical trials
1) Open the Clinical Trials page.
2) Filter by sponsor or phase.

Link market and interest
1) Open the Market & Trends page.
2) Select region and term.

3- Functional Requirements

- Show (KPI cards).
- Compare the seven drugs by serious and death rates.
- Show the top 10 reactions per drug.
- Show report trends over the years.
- Analyze reports by age group, sex and country.
- Show clinical trials by phase, status, sponsor and condition.
- Compare LLY and NVO prices over time.
- Provide interactive filters for drug, year, country, sex, manufacturer, and trial status/phase.
- Allowing users to explore charts and view updated results based on their selections.

4- Non-Functional Requirements

- **Accuracy:** figures match the source data.
- **Usability:** clear design, understandable titles, consistent colors.
- **Filters work correctly and affect all related visuals.**
- **Transparency:** cleaning steps, definitions and assumptions are documented.
- **Privacy:** data is public with no patient identifiers; no personal data displayed.

5- Business Questions

1- What are the top 10 reported adverse reactions for each drug?

2- Which drugs have the highest serious, death and hospitalization rates?

3- How did unique report counts change from 2012 to 2025, and does it relate to drug launches?

4- How are reports distributed by age and sex, and does the serious rate differ between groups?

5- Which countries report the most, and what share of reports have an unknown country?

6- How are clinical trials distributed by phase and status, and which sponsor (Novo Nordisk or Eli Lilly) has more trials?

7- Which investigational drugs (orforglipron, retatrutide, etc.) are in which phase, and how many participants are enrolled?

8- How did LLY and NVO perform since 2017, and which stock returned more?

9- Did search interest (Ozempic, Mounjaro…) rise alongside stock moves or after FDA approval dates?

10- In which regions (US, GB, IN, PK, SA, AE) is interest in each term highest?

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